

Lesson 5: Delta Hedging and Gamma Scalping

Analyst onboarding · Lesson 5 of 10 · L/S Gamma Desk, QUANTT

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Goal: see how we remove the direction bet from an option and turn it into a pure bet on movement, on both the long side (forward scalping) and the short side (reverse scalping).

You need: Lesson 4 (delta, gamma, theta).

1 Delta hedging

Delta hedging means trading SPY shares so your total delta is zero. If you buy the 761 call (+50 deltas), you **short 50 SPY shares**. Small moves in SPY now barely change your P&L.

Because of gamma, the delta drifts as SPY moves. **Re-hedging** means trading shares to bring it back to zero. **Gamma scalping** is the profit or loss from doing that, again and again.

⇒ Once hedged, the position no longer cares about **direction**, only about **how much** SPY moves.

2 Forward scalping: long gamma

You own options, so your delta moves **with** SPY. Re-hedging makes you trade **against** the move:

1. SPY goes **up** → your delta turns positive → you **sell** shares (selling high).
2. SPY goes **down** → your delta turns negative → you **buy** shares (buying low).

Example (round numbers): SPY at \$500, and your position has gamma $\Gamma = 4$ shares per \$1 (about three ATM contracts).

1. SPY rises to \$505 → delta is +20 → sell 20 shares at \$505.
2. SPY falls back to \$500 → delta is 0 again → buy 20 shares back at \$500.
3. Hedging profit: $20 \times \$5 = \100 .

Each \$5 leg earns $\frac{1}{2}\Gamma(\Delta S)^2 = \frac{1}{2}(4)(5^2) = 50$ dollars, so the round trip makes \$100.

The catch: you pay **theta** every day. If SPY sits still, you slowly bleed.

3 Reverse scalping: short gamma

You sold options, so your delta moves **against** SPY. Re-hedging makes you trade **with** the move:

1. SPY goes **up** → your delta turns negative → you **buy** shares (buying high).
2. SPY goes **down** → your delta turns positive → you **sell** shares (selling low).

The same round trip **costs** you \$100. **The payoff:** you **collect theta** every day. If SPY moves less than implied, the theta beats the hedging losses.

4 The P&L formula

Over each hedging interval Δt , a delta-hedged option earns approximately:

$$\text{P\&L} \approx \frac{1}{2} \Gamma S^2 \left(\frac{(\Delta S)^2}{S^2} - \sigma_{IV}^2 \Delta t \right)$$

- $(\Delta S)^2/S^2$ is the **realized variance**: how much SPY actually moved (variance is volatility squared).
- $\sigma_{IV}^2 \Delta t$ is the **implied variance**: how much the option was priced to move.
- $\frac{1}{2} \Gamma S^2$ is the **dollar gamma**: the size of your bet.

$\Rightarrow$ Long gamma wins when **realized** > **implied** (RV > IV). Short gamma wins when **implied** > **realized** (IV > RV).

5 The breakeven move

Setting the bracket to zero gives the daily move at which theta exactly equals the gamma gains:

$$\text{Breakeven move} \approx S \times \frac{\sigma_{IV}}{\sqrt{252}}$$

Example: SPY \$500 and IV 16% $\rightarrow$ the breakeven is $500 \times 0.16 / \sqrt{252} \approx \5.04 a day. With $\Gamma = 4$, daily theta $\approx \frac{1}{2}(4)(5.04^2) \approx \51 .

- SPY moves \$5 up and back in a day: the long-gamma trader makes \$100 from hedging against \$51 of theta. **Profit.**
- SPY moves only \$2 up and back: hedging makes $2 \times \frac{1}{2}(4)(2^2) = \16 against \$51 of theta. **Loss**, and a win for the short-gamma trader.

6 Side by side

	Forward scalping	Reverse scalping
Gamma	Long	Short
Theta	Pay	Collect
Re-hedging	Buy low, sell high	Buy high, sell low
Wins when	RV > IV	IV > RV
Best market	Large, choppy moves	Quiet, range-bound
Worst market	Flat and sleepy	Gaps and crashes
Worst loss	Premium paid	Potentially very large

7 How often to re-hedge

The formula assumes you re-hedge every Δt . In practice you choose:

1. **Time-based:** on a schedule, e.g. hourly or daily. **Our desk hedges once a day.**
2. **Band-based:** only when delta drifts past a threshold, e.g. ± 10 shares.

Hedging more often tracks the theory more closely, but every trade pays the bid-ask spread. Long gamma wants to hedge often enough to **lock in** gains. Short gamma often uses wider bands, because if SPY reverts before you hedge, you **avoid** the loss.

8 Risks to know

1. **Theta bleed (long gamma):** quiet markets cost money every day.
2. **Gap risk (short gamma):** you can't hedge overnight or through a crash, and one gap can erase weeks of theta.
3. **Vega risk (both):** the formula ignores IV changes. If IV spikes, long positions gain and short positions lose, even before any hedging.
4. **Costs (both):** spreads and commissions can turn a paper profit into a real loss.

9 In our code

- `core/Algos/Hedging/delta_hedge.py` sets the target to minus the option delta: `target_shares = -option_delta`.
- `Hedging/portfolio.py` (rebalance) trades the difference between target and current shares.
- In our smoke test, buying 1 call and selling 1 put gave about +100 deltas, and the hedger sold **101 shares**.

10 Words to know

- **Delta hedge / re-hedge:** trade shares to make delta zero / do it again after SPY moves.
- **Forward / reverse scalping:** hedging a long / short gamma position.
- **Realized / implied variance:** volatility squared, actual / priced in. **Breakeven move:** the daily move where theta equals gamma gains.

11 Check yourself

1. You are short gamma and SPY rallies. Do you buy or sell shares to re-hedge?
2. SPY \$760, IV 14%: what is the daily breakeven move?
3. Why might a short-gamma trader prefer wider hedging bands?

Answers: (1) Buy, chasing the move. (2) $760 \times 0.14 / \sqrt{252} \approx \6.70 . (3) Fewer forced “buy high, sell low” trades if SPY reverts.